

A Federated Learning Study on Bias and Fairness in Small Data Medical Applications

Pratik Korat
San Jose State University
Email: pratikkumardalsukhbhai.korat@sjsu.edu

Kirthika Ashokkumar
San Jose State University
Email: kirthika.ashokkumar@sjsu.edu

Sakina Rahman
San Jose State University
Email: sakina.rahman@sjsu.edu

Palak Agarwal
San Jose State University
Email: palak.agarwal@sjsu.edu

Mahima Agumbe Suresh
San Jose State University
Email: mahima.agumbesuresh@sjsu.edu

Abstract—Important concerns when using medical data for Machine Learning (ML) is patient privacy and bias. Federated Learning (FL), the training of a centralized model by using parameters from decentralized models, is alternatively used to protect patient privacy. Medical data can often be structured and sparse, where deep learning is not applicable. Besides, medical applications are at risk for biases in model accuracies for different populations, such as gender bias, age bias, etc. This work studies the efficacy of federated learning for small data medical applications and explore bias and fairness in these models arising from a federated learning setting.

I. INTRODUCTION

Machine Learning (ML) has accelerated our abilities to draw insights from data and make predictions. In medical applications, it carries the potential to catch diseases early. ML predictions are typically made by models that are trained on data in a centralized fashion. Collecting data in one place raises privacy concerns. To tackle this problem, Federated learning (FL) was developed, originating from Google's pioneering work [1] to train ML models in a distributed fashion and collect only model parameters centrally. FL has since found applications in several other domains [2], [3], including medical applications.

Most existing work in FL in medical applications focuses on deep learning models, and the input data is not columnar, e.g., images. However, there are several medical applications where the number of data points in each facility may be limited, making the datasets unsuitable for deep learning. In these applications, simpler machine learning models such as logistic regression, Naive Bayes Classifiers, and Support Vector Machines may be a better fit. If insights from multiple medical facilities can be aggregated into one model, the model can lead to better results than each facility running a separate model. However, aggregating the data in one place may lead to privacy breaches. This makes FL a suitable candidate.

Implementing FL for simpler models requires careful thought. First, preprocessing on the data is required in this context. Features need to be encoded, scaled, and normalized to prepare them for the ML model. This can be challenging to do without sharing data among facilities. In addition, FL assumes that the underlying ML model parameters can be aggregated and

“averaged” to form a global model. The simple ML models by default are not designed as such. Finally, with medical applications, it is important to understand how the FL approach impacts bias and fairness in the ML models.

In this work, we study an FL approach to train small columnar datasets. We specifically study the feasibility of FL and design experiments to study bias and fairness in these models. We experiment with three classical machine learning models, namely Logistic Regression, Support Vector Machines, and Naive Bayes Classifier on two datasets [4], [5]. The models are simple, defined by a small number of parameters, and less prone to overfitting.

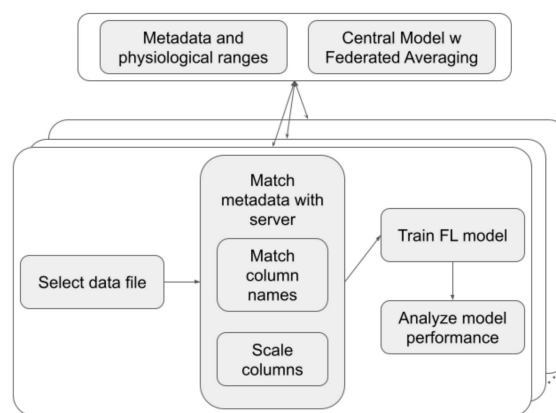


Fig. 1. Block diagram of envisioned framework

II. VISION AND BACKGROUND FOR A FEDERATED LEARNING FRAMEWORK FOR MEDICAL APPLICATIONS

The domain of medical applications brings about several engineering challenges. Each medical center, i.e., “client”, may have data in a different format. They may have skewed distributions in certain columns. For example, a birthing center would have a skewed distribution in the gender or sex column as compared to a general physician’s clinic. This makes basic preprocessing tasks such as normalization challenging.

Nonetheless, medical applications also have the advantage of domain knowledge about certain physiological parameters. For example, there is a normal range of blood pressure readings known in the field. This enables us to use domain expertise in preprocessing. Therefore, we envision a secure and easy to use system that can be used and interpreted by medical professionals easily, as depicted in Figure 1.

Studies on fairness began with ensuring that similar individuals receive similar predictions [6]. Metrics were then studied for bias and fairness in traditional ML, including equality of opportunity [7]. However, fairness in the context of FL has not received as much attention. A recent survey shows that with FL, data heterogeneity among clients can reduce model performance and worsens fairness and bias issues [8]. This motivates the need for a study to quantify bias in FL settings and study the factors leading to or limiting bias and fairness.

III. EXPERIMENTS

In this section, we detail the datasets, models, and metrics used in this study.

A. Datasets

We selected two datasets from the thousands of available datasets in the medical domain. The first one is for diabetes risk prediction [5] and the second for heart disease [4]. The diabetes dataset comprises 120 instances of 16 different attributes related to diabetes diagnosis. The heart disease dataset comprises of 921 instances of 15 different attributes.

The datasets required preprocessing including encoding, scaling, and normalization. To prepare them for the FL framework, we split the datasets into different clients and a server test dataset. These splits were performed before preprocessing to avoid data snooping. For each of the experiments details in the next subsections, 8 experiments were setup with different splits that were not independent and identically distributed.

B. Models

Our experiments were conducted on three simple models as described below.

1) *Logistic Regression*: Logistic Regression (LR) is a widely linear model. It uses the sigmoid function $\theta(s) = \frac{1}{1+e^{-s}}$ on a “signal” obtained as a linear combination, i.e., a weighted sum of the input features. The learning algorithm tunes the weights based on the data by minimizes the errors it makes on the training dataset. In the FL setting, we average these weights by training each client separately on their dataset to obtain the FL model.

2) *Gaussian Naive Bayes Classifier*: Gaussian Naive Bayes (GNB) follows a simple premise that each of the input features can independently predict the output. Each input feature is assumed to have a Gaussian distribution with a mean μ and standard deviation σ . The model finds the μ s and σ s for each feature and uses the Gaussian distribution formulation to compute the probability of belonging to class for a given sample, i.e., $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}(\frac{x-\mu}{\sigma})^2}$. For the FL setting, we average the means and standard deviations of each client.

3) *Support Vector Machines*: Support Vector Machines (SVM) are models that essentially solve an optimization problem to maximize the distance of the hyperplane separating the classes in the data from data points in each class. When using linear kernels, the parameters learned through training are similar to LR. For the FL setting, we take average of the weights from each client.

C. Metrics of Performance Evaluation and Bias/Fairness

To measure the performance of each model against both datasets, we use the accuracy metric. Accuracy is defined as $\frac{TP+TN}{TP+TN+FP+FN}$, where TP is the number of true positive, TN is the number of true negative, FP is the number of false positive, and FN is the number of false negative predictions. In previous work [9] on the diabetes dataset, it was shown that the performance of FL is better than what individual clients can independently achieve on an average. We validate the result in this paper by comparing the average of 8 experiment, each performing three rounds of FL training.

To study bias and fairness, we choose the sensitive column in the dataset to be the “sex” column containing male and female patients. We evaluate our models on the following metrics:

1) *Demographic Disparity Difference*: The demographic disparity difference (DDD) in our experiments is measured as the absolute difference in the selection rate, i.e., the probability of predicting a positive outcome, between the male and female patients. For example, if the model predicts a positive outcome for 80% of male patients and 60% of female patients, the DDD is 0.2. The closer this number is to 0, the better.

2) *Demographic Disparity Ratio*: The demographic disparity ratio (DDR) in our experiments is measured as the ratio of the selection rate, i.e., the probability of predicting a positive outcome, between the minimum of male and female patients and the maximum. For example, if the model predicts a positive outcome for 80% of male patients and 60% of female patients, the DDR is $0.6/0.8 = 0.75$. The closer this number is to 1, the better.

3) *Equalized Odds Difference*: The equalized odds difference (EOddD) in our experiments is the highest of the difference in true positive rates and false positive rates between male and female patients. The closer this value is to 0, the better.

4) *Equalized Opportunity Difference*: The equalized opportunity difference (EOppD) in our experiments is the difference between true positive rates between male and female patients. The closer this value is to 0, the better.

D. Experimental Setup

Our experiments were run on Google Cloud Platform’s Python 3 Google Compute Engine. For federated learning, we use the flower library [10]. To compute fairness metrics, we use FairLearn [11].

IV. RESULTS AND DISCUSSIONS

In this section, we present the results of our experiments.

Figure 9 depicts the results of experimentation with logistic regression on both datasets. Overall, the results show that the

Results for heart dataset using Gaussian Naive Bayes Classifier

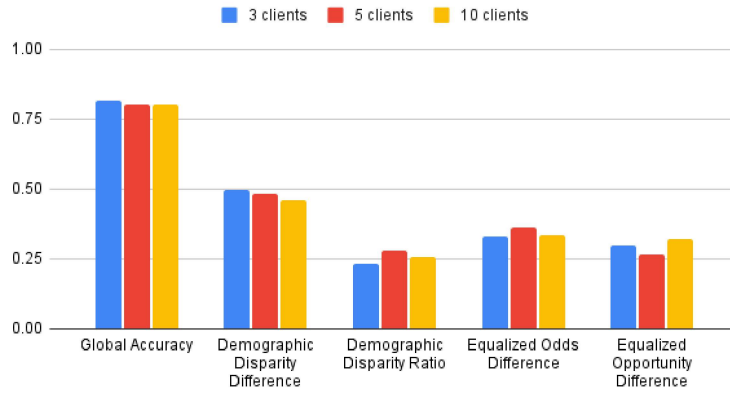


Fig. 2. Results of experimentation with Gaussian Naive Bayes Classifier on the Heart disease dataset

3 clients

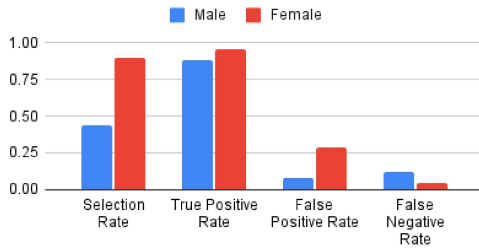


Fig. 3. Fairness metrics for 3 clients

10 clients

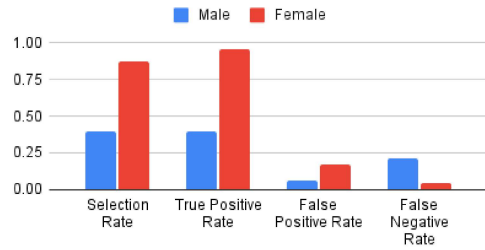


Fig. 5. Fairness metrics for 10 clients

5 clients

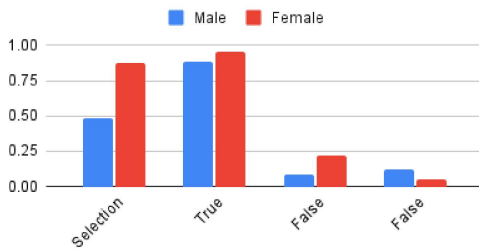


Fig. 4. Fairness metrics for 5 clients

3 clients

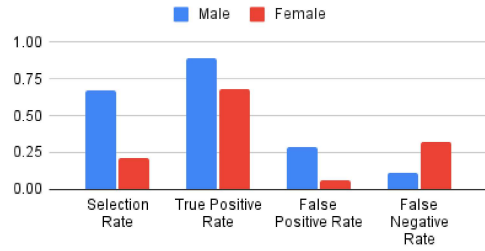


Fig. 6. Fairness metrics for 3 clients on the heart dataset

accuracy of the global model, i.e., using FL, is not impacted by the number of clients. There is demographic disparity difference and ratios show that there is a bias. To understand this further, we plotted the selection rate, false positive rate, true positive rate, and false negative rates on the diabetes dataset. Results are shown for the diabetes dataset in figures 3, 4, and 5. The graphs show similar trends for each experiment. Results for the heart disease dataset are shown in figures 6, 7, and 8. The graphs show similar trends for each experiment.

We further experimented with the heart dataset using the GNB and SVM classifiers. The results are depicted in figures 2 and 10.

V. CONCLUSIONS

This paper presents a study of bias and fairness on a federated learning framework for medical applications with small data. The models we experimented with are simple and generalizable.

5 clients

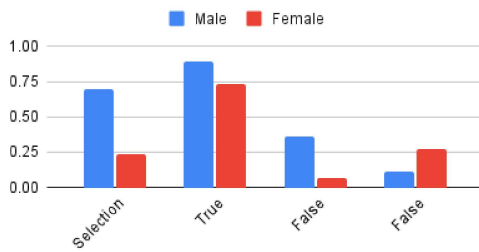


Fig. 7. Fairness metrics for 5 clients on the heart dataset

10 clients

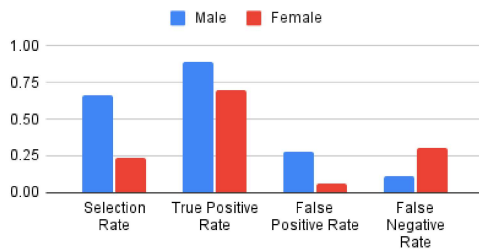


Fig. 8. Fairness metrics for 10 clients on the heart dataset

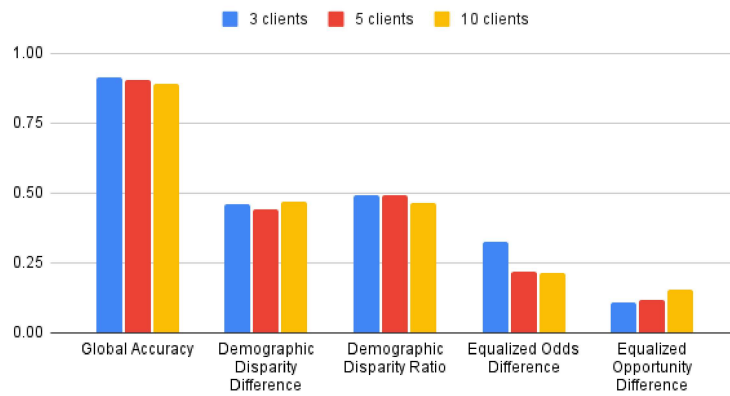
The results show that there is bias in the models. Exploring the true cause of this bias is left for future work.

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Results for Diabetes dataset using Logistic Regression



Results for Heart dataset using Logistic Regression

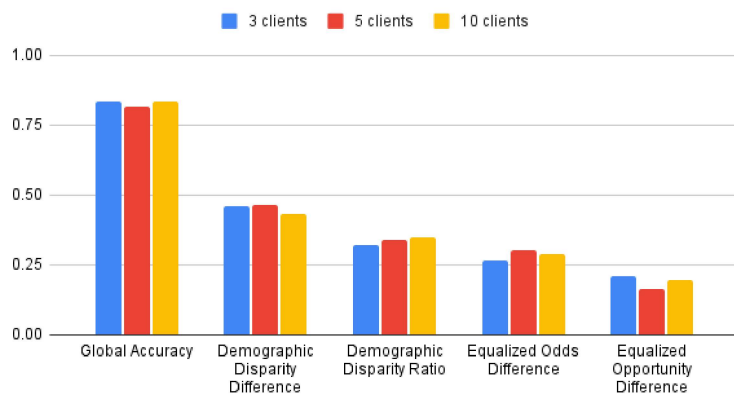


Fig. 9. Results of experimentation with Logistic Regression on the Diabetes dataset and the Heart disease dataset

Results for heart dataset using Support Vector Classifier

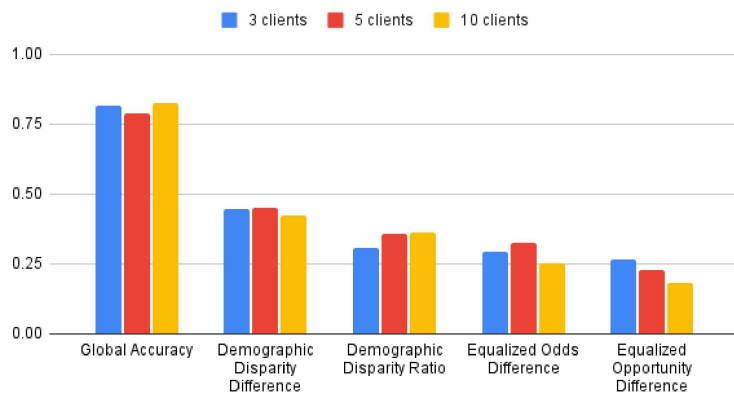


Fig. 10. Results of experimentation with Support Vector Machine Classifier on the Heart disease dataset